

Lab 9: Gravimetric Analysis of Chloride

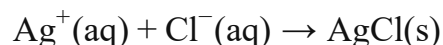
Chemistry 150 - Winter 2026

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Unknown #	[unknown number]

Introduction

Gravimetric analysis is a method of determining the amount of a target species by converting it into a solid compound, filtering and drying that solid, and weighing it precisely. Because the solid has a known chemical formula, the mass of the target can be back-calculated from the mass of the precipitate.

In this experiment, the target is chloride ion (Cl^-) dissolved in an unknown sample. When silver nitrate solution (AgNO_3) is added, Ag^+ reacts with Cl^- to form insoluble silver chloride:



The extent to which this precipitation is complete is governed by the solubility product of AgCl :

$$K_{sp} = [\text{Ag}^+][\text{Cl}^-] = 1.8 \times 10^{-10}$$

If Ag^+ is kept in large excess, the equilibrium forces $[\text{Cl}^-]$ to an extremely low value, ensuring virtually all chloride is captured as precipitate.

AgCl initially forms as a colloid — particles too small to be caught by a filter. Heating the mixture and adding electrolyte (HNO_3) removes the surface charges that keep the particles apart, causing them to clump into larger particles that can be filtered. AgCl is sensitive to light: exposure reduces Ag^+ to grey or black Ag metal, so the mixture is stored in a dark cupboard during aging.

Objective

Determine the mass percentage of chloride (Cl^-) in an unknown sample using gravimetric precipitation as AgCl , and report the average result from two independent trials.

Method

The procedure followed Lab 9 of the Chemistry 150 Lab Manual W 2026. Two independent trials were performed.

For each trial, approximately 0.4 g of the unknown was weighed precisely into a 400 mL beaker. 200 mL of deionised water and 5 drops of 6 M HNO₃ were added. 5% (w/v) AgNO₃ solution was added slowly with stirring until the precipitate began to coagulate, then 5 mL more was added. The mixture was heated to near boiling and simmered for a few minutes until the solution cleared. A few drops of AgNO₃ were added to confirm no further precipitation. The beaker was then placed in a dark cupboard for 30 minutes.

A pre-weighed filter paper was prepared. The supernatant was decanted through the filter, the precipitate washed into the filter with small amounts of 0.03 M HNO₃, then rinsed with approximately 15 mL of deionised water. The filter was spread on a watch glass and dried overnight in the oven. The dried filter and precipitate were weighed the following day.

Trial 2 was started while Trial 1 was aging in the dark cupboard.

Data

Table 1. Raw measurements for both trials.

Trial	Unknown mass (g)	Filter mass, initial (g)	Filter mass, final (g)	Mass of AgCl (g)
1				
2				

Results

The mass of AgCl precipitate for each trial is:

$$m_{\text{AgCl}} = m_{\text{filter, final}} - m_{\text{filter, initial}}$$

Since Ag⁺ and Cl⁻ react in a 1:1 ratio, moles of AgCl equals moles of Cl⁻:

$$n_{\text{Cl}^-} = n_{\text{AgCl}} = \frac{m_{\text{AgCl}}}{M_{\text{AgCl}}} = \frac{m_{\text{AgCl}}}{143.32 \text{ g/mol}}$$

The mass of Cl⁻ in the sample:

$$m_{\text{Cl}^-} = n_{\text{Cl}^-} \times 35.45 \text{ g/mol}$$

The percentage of Cl^- in the unknown:

$$\% \text{Cl}^- = \frac{m_{\text{Cl}^-}}{m_{\text{unknown}}} \times 100\%$$

Combining into one expression:

$$\% \text{Cl}^- = \frac{m_{\text{AgCl}} \times 35.45}{143.32 \times m_{\text{unknown}}} \times 100\%$$

Table 2. Calculated results for both trials.

Trial	Mass of AgCl (g)	Moles of Cl^- (mol)	Mass of Cl^- (g)	% Cl^-
1				
2				

Average % Cl^- :

$$\overline{\% \text{Cl}^-} = \frac{\text{Trial 1} + \text{Trial 2}}{2} =$$

Discussion

Ensuring complete precipitation. AgNO_3 was added in excess beyond the point where precipitation stopped. According to the K_{sp} expression, a large excess of Ag^+ drives $[\text{Cl}^-]$ extremely low — on the order of 10^{-8} mol/L — so the amount of Cl^- left in solution is negligible compared to the amount precipitated.

Coagulation. AgCl initially forms a colloid that passes through filter paper. Heating and the presence of HNO_3 as electrolyte neutralise the surface charges on the colloidal particles, allowing them to coalesce into filterable clumps. Simmering until the solution cleared was used as the visual indicator that coagulation was complete.

Light sensitivity. The mixture was kept in a dark cupboard during aging and handled quickly during filtering. Any purple or grey discolouration of the precipitate would indicate partial reduction of AgCl to Ag metal, which would add mass and give a high result for % Cl^- .

Agreement between trials. [Comment on how close Trial 1 and Trial 2 results are. If within ~1%, the method is working well. If they differ more, consider whether a transfer loss or light exposure affected one trial.]

Sources of error. The main sources of error are: incomplete transfer of precipitate from the beaker to the filter (leads to a low result), co-precipitation of AgNO_3 trapped in the colloid (leads to a high result), and light exposure darkening the precipitate (leads to a high result). Using the rubber policeman only as a last resort and minimising light exposure reduces these effects.

Conclusion

The percentage of Cl^- in the unknown (# [unknown number]) was determined by precipitating it as AgCl and weighing the dried precipitate. Trial 1 gave []% and Trial 2 gave []%, for an average of []% Cl^- . The two trials agreed to within []%, indicating [good/acceptable] reproducibility.

Reference

Chemistry 150 Lab Manual W 2026, Lab 9: Gravimetric Analysis of Chloride.